Comp2151 – Agile Software Development

Group Project – ABC Recycling Management Inc.

# Project Report

## Group 49


**Team Members**

Ceejay Dawe | Student ID: 101121107

Brent Joshua Samson | Student ID: 101580271


**Instructor / Product Owner:**

Abid Rana, Operations Manager – ABC Recycling Management Inc.

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# 1. Product Vision Statement

FOR citizens and municipalities in Canadian cities WHO need reliable, transparent, and efficient waste collection services, THE ABC Recycling Management System IS a web-based operations and citizen engagement platform THAT enables ABC Recycling to schedule routes, manage fleet and personnel, process service requests, and proactively notify citizens about pickups and service updates.

UNLIKE manual phone-based systems or disconnected spreadsheets, OUR PRODUCT provides an integrated, real-time platform that improves operational efficiency, enhances citizen satisfaction, and supports ABC's growth as it wins additional municipal contracts across Canada.

| Vision Component | Description |
|---|---|
| **For (Target Customer)** | Canadian city residents, and ABC Recycling operations/office staff |
| **Who (Need)** | Need a transparent, reliable, and self-service waste management experience |
| **The (Product)** | ABC Recycling Management System |
| **Is a (Category)** | Web-based operations and citizen portal platform |
| **That (Key Benefit)** | Manages schedules, notifications, complaints, containers, fleet GPS, and fuel reconciliation in one system |
| **Unlike (Alternative)** | Manual, phone-based, or disconnected spreadsheet-driven processes |
| **Our Product (Differentiator)** | Integrates real-time GPS tracking, automated citizen notifications, and fuel voucher reconciliation into a single platform |

# 2. Business Case

## 2.1 Executive Summary

ABC Recycling Management Inc. has been awarded a garbage and recycling collection contract by a major Canadian city, with operations commencing January 1, 2027. To support these operations, ABC requires a purpose-built software system that manages pickup scheduling, fleet operations, citizen engagement, container inventory, and fuel cost reconciliation. Without this system, ABC cannot scale its operations efficiently or meet the transparency expectations of the city and its residents. The system is targeted for delivery by December 1, 2026.

## 2.2 Recommended Solution

We recommend developing a custom web-based system with two distinct portals: (1) an internal operations portal for ABC staff (scheduling, fleet management, complaint handling, fuel reconciliation) and (2) a citizen-facing portal (account registration, notifications, complaints, container requests, status tracking). This approach was selected because:

- No off-the-shelf waste management software meets all of ABC's specific requirements (GPS integration, voucher reconciliation, custom zoning).
- A custom web application can be designed to support future contract expansions to other cities.
- Web-only delivery (no mobile app) is confirmed by the client, reducing scope and cost.

## 2.3 Implementation Plan

The project will follow a SCRUM Agile framework with iterative sprints. High-level milestones are:

| Phase | Milestone | Target Date |
|---|---|---|
| 1 | Requirements Gathering Complete | February 28, 2026 |
| 2 | System Design & Architecture | March 31, 2026 |
| 3 | Core Module Development (Sprints 1–4) | April – July 2026 |
| 4 | Integration & Testing | August – October 2026 |
| 5 | UAT & Client Sign-Off | November 2026 |
| 6 | Go-Live | December 1, 2026 |

## 2.4 Risk Assessment

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| GPS integration complexity | Medium | High | Spike/proof-of-concept in Sprint 1 |
| Scope creep from new city contracts | High | Medium | Strict change control process; modular architecture |
| Delayed city data (zones, routes) | Medium | High | Use placeholder data in testing; admin setup tool |
| Staff adoption/training | Medium | Medium | Conduct UAT with actual staff; provide training docs |
| SMS/Email gateway failure | Low | High | Use failover email provider; log all notification attempts |

## 2.5 Stakeholder Analysis

| Stakeholder | Role | Interest | Influence |
|---|---|---|---|
| John Doe (CEO) | Executive Sponsor | High | High |
| Abid Rana (Ops Manager) | Product Owner / IS Liaison | High | High |
| Office Staff | Primary end users | High | Medium |
| Truck Crew | Route execution | Low | Low |
| Technicians | Maintenance & delivery | Medium | Low |
| Citizens | Portal users | Medium | Low |
| Gas Station Chain | Fuel vendor / invoicer | Low | Low |
| City Government | Contract authority | High | High |

## 2.6 Conclusion

The ABC Recycling Management System is a mission-critical investment that enables ABC to fulfill its municipal contract, deliver excellent citizen service, and position itself competitively for future contracts. The recommended custom web platform offers the best balance of

functionality, scalability, and cost. We recommend immediate approval of this project to meet the December 1, 2026 go-live deadline.

# 3. Agile Team Charter

## 3.1 Team Identity

Team Name: Group 49 – RecycleDevs

Project: ABC Recycling Management System

Course: Comp2151 – Agile Software Development, George Brown College

## 3.2 Team Members & Roles

| Name | SCRUM Role | Responsibilities |
|---|---|---|
| Ceejay Dawe | SCRUM Master / Requirements Lead | Facilitate sprints, conduct client interviews, write user stories, build portal UI |
| Brent Joshua Samson | Developer / Tester | Build backend logic, write acceptance criteria, design wireframes, test features |

## 3.3 Team Norms & Working Agreements

- Meetings held weekly (online via Teams) and during dedicated class time
- All decisions are made by consensus; unresolved issues escalated to the SCRUM Master
- Code and documents committed to shared repository within 24 hours of completion
- All team members respond to messages within 24 hours
- Each member attends scheduled meetings or provides advance notice of absence
- No one completes another member's work without agreement; pair-programming encouraged

## 3.4 Definition of Done

- User story implemented and peer-reviewed
- Acceptance criteria verified by tester
- Wireframe or screen included in report if applicable
- Section of report written and spell-checked

# 4. User Personas

Based on the System Service Request and the client interview conducted February 20, 2026, the following personas represent the key user categories of the ABC Recycling Management System.

## 4.1 Citizen Personas

### Persona 1.1: Maria Chen

| | |
|---|---|
| **Name** | Maria Chen |
| **Age** | 42 |
| **User Type** | Citizen |
| **Profile** | Homeowner, working professional, registers for pickup reminders to keep her schedule organized. Frustrated by missed pickups and wants fast resolution online without making phone calls. |

### Persona 1.2: James Okafor

| | |
|---|---|
| **Name** | James Okafor |
| **Age** | 67 |
| **User Type** | Citizen |
| **Profile** | Retired resident, prefers email over SMS. Needs an easy-to-use portal to report a damaged bin. Values clear status updates so he knows someone is handling his issue. |

### Persona 1.3: Priya Sharma

| | |
|---|---|
| **Name** | Priya Sharma |
| **Age** | 29 |
| **User Type** | Citizen |
| **Profile** | Renter in a multi-unit building. Multiple residents share the address. Registers for text notifications. Often files missed pickup complaints and monitors resolution status. |

## 4.2 Office Staff Personas

### Persona 2.1: Sandra Kowalski

| | |
|---|---|
| **Name** | Sandra Kowalski |
| **Age** | 38 |
| **User Type** | Office Staff |
| **Profile** | Customer-facing assistant who handles incoming complaints and maintenance requests. Needs a clear dashboard to assign, track, and close cases quickly. |

### Persona 2.2: Derek Nguyen

| | |
|---|---|
| **Name** | Derek Nguyen |
| **Age** | 45 |
| **User Type** | Office Staff |
| **Profile** | Scheduling coordinator who manages pickup routes, zone assignments, and crew scheduling. Needs the interactive map and scheduling tools to stay on top of daily operations. |

### Persona 2.3: Fatima Al-Hassan

| | |
|---|---|
| **Name** | Fatima Al-Hassan |
| **Age** | 33 |
| **User Type** | Office Staff |
| **Profile** | Invoicing assistant responsible for entering fuel vouchers daily and reconciling the gas station invoice every 15 days. Detail-oriented, needs clear mismatch alerts. |

## 4.3 Technician Personas

### Persona 3.1: Carlos Rivera

| | |
|---|---|
| **Name** | Carlos Rivera |
| **Age** | 31 |
| **User Type** | Technician |

| | |
|---|---|
| **Profile** | Field technician who handles container repairs and replacements. Works from a mobile-friendly job list. Needs to update job status in the field and collect payment on delivery. |

### Persona 3.2: Tony Barzetti

| | |
|---|---|
| **Name** | Tony Barzetti |
| **Age** | 48 |
| **User Type** | Technician |
| **Profile** | Senior technician handling complex repairs. Prefers a clear view of serial numbers and container history so he arrives prepared. |

### Persona 3.3: Aisha Diallo

| | |
|---|---|
| **Name** | Aisha Diallo |
| **Age** | 26 |
| **User Type** | Technician |
| **Profile** | Junior technician new to the job. Relies on the job ticket details (address, container type, issue description) to complete assignments correctly without supervisor guidance. |

## 4.4 Manager Personas

### Persona 4.1: Abid Rana

| | |
|---|---|
| **Name** | Abid Rana |
| **Age** | 52 |
| **User Type** | Manager |
| **Profile** | Operations Manager and primary system user. Needs full system access — route management, GPS tracking, complaint dashboard, fuel invoice approval, and staff management. Acts as the single point of contact for escalations. |

### Persona 4.2: Lynn Tran

| | |
|---|---|
| **Name** | Lynn Tran |

| | |
|---|---|
| **Age** | 44 |
| **User Type** | Manager |
| **Profile** | Assistant Manager who oversees daily operations when Abid is unavailable. Needs read access to all modules and write access to scheduling and complaints. |

### Persona 4.3: John Doe

| | |
|---|---|
| **Name** | John Doe |
| **Age** | 58 |
| **User Type** | Manager |
| **Profile** | CEO and executive sponsor. Rarely logs in, but expects a high-level financial and operations dashboard to monitor contract compliance and service quality KPIs. |

# 5. User Stories

The following user stories were derived from the System Service Request and the requirements interview with the Product Owner (Abid Rana) on February 20, 2026.

| ID | Role | User Story | Priority |
|---|---|---|---|
| US-01 | Citizen | As a Citizen, I want to register an online account so that I can receive pickup notifications and track my service requests. | Must Have |
| US-02 | Citizen | As a Citizen, I want to verify my email address so that my account is activated securely. | Must Have |
| US-03 | Citizen | As a Citizen, I want to receive email and/or text reminders about upcoming garbage, recycling, and organics pickups so that I never miss a collection day. | Must Have |
| US-04 | Citizen | As a Citizen, I want to submit a missed pickup complaint online so that ABC can schedule a special pickup the next day. | Must Have |
| US-05 | Citizen | As a Citizen, I want to track the status of my complaint through the online portal so that I know when my issue will be resolved. | Must Have |
| US-06 | Citizen | As a Citizen, I want to submit a container maintenance request online so that a technician can be scheduled to repair or replace my bin. | Must Have |
| US-07 | Citizen | As a Citizen, I want to request a replacement container and pay a $40 fee (cash or check on delivery) so that I can get a new bin when mine is damaged beyond repair. | Must Have |
| US-08 | Office Staff | As a Office Staff, I want to view and manage citizen complaints so that I can assign and resolve missed pickup cases efficiently. | Must Have |
| US-09 | Office Staff | As a Office Staff, I want to schedule pickup routes by zone and day of the week so that all neighborhoods are served on the correct schedule. | Must Have |
| US-10 | Office Staff | As a Office Staff, I want to manage zones and routes on an interactive map so that I can draw, edit, and redistribute streets between routes as needed. | Must Have |

| | | | |
|---|---|---|---|
| US-11 | Office Staff | As a Office Staff, I want to assign drivers and helpers to specific routes so that every scheduled route has the correct crew. | Must Have |
| US-12 | Office Staff | As a Office Staff, I want to reschedule or cancel pickups due to holidays or emergencies so that the schedule remains accurate and citizens are notified. | Must Have |
| US-13 | Office Staff / Invoicing | As a Office Staff / Invoicing, I want to enter fuel voucher details into the system so that we maintain an internal record of all gas purchases by truck. | Must Have |
| US-14 | Office Staff / Invoicing | As a Office Staff / Invoicing, I want the system to reconcile the gas station invoice against entered vouchers so that discrepancies are highlighted for investigation. | Must Have |
| US-15 | Manager | As a Manager, I want to view a dashboard showing missed pickup cases and their resolution status so that I can monitor service quality. | Must Have |
| US-16 | Manager | As a Manager, I want to view a financial dashboard showing fuel invoices and voucher reconciliation so that I can approve payments to the gas station. | Should Have |
| US-17 | Manager | As a Manager, I want to track fleet trucks in real time on a city map via GPS so that I can monitor all vehicles during operations. | Must Have |
| US-18 | Technician | As a Technician, I want to view my scheduled maintenance and container delivery jobs so that I can plan and complete my daily assignments. | Must Have |
| US-19 | Technician | As a Technician, I want to update the status of a maintenance or replacement job (e.g., In Progress, Completed) so that the system and citizen portal reflect the current state. | Must Have |
| US-20 | Manager | As a Manager, I want to manage container inventory by serial number so that I can track which containers have been issued, repaired, or replaced at each address. | Should Have |

# 6. Story Priority Estimation

## 6.1 Business Value-Addition & MoSCoW

| ID | Story Summary | Business Value (0–100) | MoSCoW | Rationale |
|---|---|---|---|---|
| US-01 | As a Citizen, I want to register an online account so that I can receive pickup notifications and track my service requests. | 100 | M | Core operational need |
| US-02 | As a Citizen, I want to verify my email address so that my account is activated securely. | 80 | M | Core operational need |
| US-03 | As a Citizen, I want to receive email and/or text reminders about upcoming garbage, recycling, and organics pickups so that I never miss a collection day. | 90 | M | Core operational need |
| US-04 | As a Citizen, I want to submit a missed pickup complaint online so that ABC can schedule a special pickup the next day. | 95 | M | Core operational need |
| US-05 | As a Citizen, I want to track the status of my complaint through the online portal so that I know when my issue will be resolved. | 85 | M | Core operational need |
| US-06 | As a Citizen, I want to submit a container maintenance request online so that a technician can be scheduled to repair or replace my bin. | 80 | M | Core operational need |
| US-07 | As a Citizen, I want to request a replacement container and pay a $40 fee (cash or check on delivery) so that I can get a new bin when mine is damaged beyond repair. | 75 | M | Core operational need |
| US-08 | As a Office Staff, I want to view and manage citizen complaints so that I can assign and resolve missed pickup cases efficiently. | 90 | M | Core operational need |
| US-09 | As a Office Staff, I want to schedule pickup routes by zone and day of the week so that all neighborhoods are served on the correct schedule. | 100 | M | Core operational need |

| | | | | |
|---|---|---|---|---|
| US-10 | As a Office Staff, I want to manage zones and routes on an interactive map so that I can draw, edit, and redistribute streets between routes as needed. | 95 | M | Core operational need |
| US-11 | As a Office Staff, I want to assign drivers and helpers to specific routes so that every scheduled route has the correct crew. | 90 | M | Core operational need |
| US-12 | As a Office Staff, I want to reschedule or cancel pickups due to holidays or emergencies so that the schedule remains accurate and citizens are notified. | 85 | M | Core operational need |
| US-13 | As a Office Staff / Invoicing, I want to enter fuel voucher details into the system so that we maintain an internal record of all gas purchases by truck. | 80 | M | Core operational need |
| US-14 | As a Office Staff / Invoicing, I want the system to reconcile the gas station invoice against entered vouchers so that discrepancies are highlighted for investigation. | 85 | M | Core operational need |
| US-15 | As a Manager, I want to view a dashboard showing missed pickup cases and their resolution status so that I can monitor service quality. | 85 | M | Core operational need |
| US-16 | As a Manager, I want to view a financial dashboard showing fuel invoices and voucher reconciliation so that I can approve payments to the gas station. | 75 | S | Valuable but not launch-critical |
| US-17 | As a Manager, I want to track fleet trucks in real time on a city map via GPS so that I can monitor all vehicles during operations. | 80 | M | Core operational need |
| US-18 | As a Technician, I want to view my scheduled maintenance and container delivery jobs so that I can plan and complete my daily assignments. | 80 | M | Core operational need |
| US-19 | As a Technician, I want to update the status of a maintenance or replacement job (e.g., In Progress, Completed) so that the system and citizen portal reflect the current state. | 85 | M | Core operational need |

| | | | | |
|---|---|---|---|---|
| US-20 | As a Manager, I want to manage container inventory by serial number so that I can track which containers have been issued, repaired, or replaced at each address. | 70 | S | Valuable but not launch-critical |

# 7. Story Point Estimation

## 7.1 SCRUM Poker & T-Shirt Sizes

Story points were estimated using Planning Poker (Fibonacci scale: 1,2,3,5,8,13,21) and T-Shirt sizes (XS, S, M, L, XL). Values represent team consensus after discussion.

| ID | User Story (Full) | SCRUM Poker (Fibonacci) | T-Shirt Size |
|---|---|---|---|
| US-01 | As a Citizen, I want to register an online account so that I can receive pickup notifications and track my service requests. | 5 | M |
| US-02 | As a Citizen, I want to verify my email address so that my account is activated securely. | 3 | S |
| US-03 | As a Citizen, I want to receive email and/or text reminders about upcoming garbage, recycling, and organics pickups so that I never miss a collection day. | 5 | M |
| US-04 | As a Citizen, I want to submit a missed pickup complaint online so that ABC can schedule a special pickup the next day. | 5 | M |
| US-05 | As a Citizen, I want to track the status of my complaint through the online portal so that I know when my issue will be resolved. | 5 | M |
| US-06 | As a Citizen, I want to submit a container maintenance request online so that a technician can be scheduled to repair or replace my bin. | 5 | M |
| US-07 | As a Citizen, I want to request a replacement container and pay a $40 fee (cash or check on delivery) so that I can get a new bin when mine is damaged beyond repair. | 3 | S |
| US-08 | As a Office Staff, I want to view and manage citizen complaints so that I can assign and resolve missed pickup cases efficiently. | 8 | L |
| US-09 | As a Office Staff, I want to schedule pickup routes by zone and day of the week so that all neighborhoods are served on the correct schedule. | 13 | XL |

| | | | |
|---|---|---|---|
| US-10 | As a Office Staff, I want to manage zones and routes on an interactive map so that I can draw, edit, and redistribute streets between routes as needed. | 13 | XL |
| US-11 | As a Office Staff, I want to assign drivers and helpers to specific routes so that every scheduled route has the correct crew. | 8 | L |
| US-12 | As a Office Staff, I want to reschedule or cancel pickups due to holidays or emergencies so that the schedule remains accurate and citizens are notified. | 8 | L |
| US-13 | As a Office Staff / Invoicing, I want to enter fuel voucher details into the system so that we maintain an internal record of all gas purchases by truck. | 5 | M |
| US-14 | As a Office Staff / Invoicing, I want the system to reconcile the gas station invoice against entered vouchers so that discrepancies are highlighted for investigation. | 8 | L |
| US-15 | As a Manager, I want to view a dashboard showing missed pickup cases and their resolution status so that I can monitor service quality. | 8 | L |
| US-16 | As a Manager, I want to view a financial dashboard showing fuel invoices and voucher reconciliation so that I can approve payments to the gas station. | 8 | L |
| US-17 | As a Manager, I want to track fleet trucks in real time on a city map via GPS so that I can monitor all vehicles during operations. | 13 | XL |
| US-18 | As a Technician, I want to view my scheduled maintenance and container delivery jobs so that I can plan and complete my daily assignments. | 5 | M |
| US-19 | As a Technician, I want to update the status of a maintenance or replacement job (e.g., In Progress, Completed) so that the system and citizen portal reflect the current state. | 5 | M |
| US-20 | As a Manager, I want to manage container inventory by serial number so that I can track which containers have been issued, repaired, or replaced at each address. | 8 | L |

# 8. Prioritized Product Backlog

The backlog is sorted by MoSCoW priority, then by Business Value (descending). Story points are from SCRUM Poker estimates.

| Rank | User Story (Full) | Role | BV | MoSCoW | Points | T-Shirt |
|---|---|---|---|---|---|---|
| 1 | [US-01] As a Citizen, I want to register an online account so that I can receive pickup notifications and track my service requests. | Citizen | 100 | M | 5 | M |
| 2 | [US-09] As a Office Staff, I want to schedule pickup routes by zone and day of the week so that all neighborhoods are served on the correct schedule. | Office Staff | 100 | M | 13 | XL |
| 3 | [US-04] As a Citizen, I want to submit a missed pickup complaint online so that ABC can schedule a special pickup the next day. | Citizen | 95 | M | 5 | M |
| 4 | [US-10] As a Office Staff, I want to manage zones and routes on an interactive map so that I can draw, edit, and redistribute streets between routes as needed. | Office Staff | 95 | M | 13 | XL |
| 5 | [US-03] As a Citizen, I want to receive email and/or text reminders about upcoming garbage, recycling, and organics pickups so that I never miss a collection day. | Citizen | 90 | M | 5 | M |
| 6 | [US-08] As a Office Staff, I want to view and manage citizen complaints so that I can assign and resolve missed pickup cases efficiently. | Office Staff | 90 | M | 8 | L |
| 7 | [US-11] As a Office Staff, I want to assign drivers and helpers to specific routes so that every | Office Staff | 90 | M | 8 | L |

| | | | | | | |
|---|---|---|---|---|---|---|
| | scheduled route has the correct crew. | | | | | |
| 8 | [US-05] As a Citizen, I want to track the status of my complaint through the online portal so that I know when my issue will be resolved. | Citizen | 85 | M | 5 | M |
| 9 | [US-12] As a Office Staff, I want to reschedule or cancel pickups due to holidays or emergencies so that the schedule remains accurate and citizens are notified. | Office Staff | 85 | M | 8 | L |
| 10 | [US-14] As a Office Staff / Invoicing, I want the system to reconcile the gas station invoice against entered vouchers so that discrepancies are highlighted for investigation. | Office Staff / Invoicing | 85 | M | 8 | L |
| 11 | [US-15] As a Manager, I want to view a dashboard showing missed pickup cases and their resolution status so that I can monitor service quality. | Manager | 85 | M | 8 | L |
| 12 | [US-19] As a Technician, I want to update the status of a maintenance or replacement job (e.g., In Progress, Completed) so that the system and citizen portal reflect the current state. | Technician | 85 | M | 5 | M |
| 13 | [US-02] As a Citizen, I want to verify my email address so that my account is activated securely. | Citizen | 80 | M | 3 | S |
| 14 | [US-06] As a Citizen, I want to submit a container maintenance request online so that a technician can be scheduled to repair or replace my bin. | Citizen | 80 | M | 5 | M |
| 15 | [US-13] As a Office Staff / Invoicing, I want to enter fuel voucher details into the system so that we | Office Staff / Invoicing | 80 | M | 5 | M |

| | | | | | | |
|---|---|---|---|---|---|---|
| | maintain an internal record of all gas purchases by truck. | | | | | |
| 16 | [US-17] As a Manager, I want to track fleet trucks in real time on a city map via GPS so that I can monitor all vehicles during operations. | Manager | 80 | M | 13 | XL |
| 17 | [US-18] As a Technician, I want to view my scheduled maintenance and container delivery jobs so that I can plan and complete my daily assignments. | Technician | 80 | M | 5 | M |
| 18 | [US-07] As a Citizen, I want to request a replacement container and pay a $40 fee (cash or check on delivery) so that I can get a new bin when mine is damaged beyond repair. | Citizen | 75 | M | 3 | S |
| 19 | [US-16] As a Manager, I want to view a financial dashboard showing fuel invoices and voucher reconciliation so that I can approve payments to the gas station. | Manager | 75 | S | 8 | L |
| 20 | [US-20] As a Manager, I want to manage container inventory by serial number so that I can track which containers have been issued, repaired, or replaced at each address. | Manager | 70 | S | 8 | L |

# 9. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for each user story to be considered complete.

### US-01 – Acceptance Criteria

- System accepts name, address, email, and optional phone number
- Duplicate emails are rejected with a clear error message
- Account remains inactive until email is verified

### US-02 – Acceptance Criteria

- System sends a confirmation email within 2 minutes of registration
- Clicking the link activates the account
- Link expires after 24 hours

### US-03 – Acceptance Criteria

- Notifications sent at least 1 day before scheduled pickup
- Citizen can choose email, SMS, or both
- Holiday reschedule triggers updated notification

### US-04 – Acceptance Criteria

- Complaint form is accessible from citizen portal without re-login
- Submission generates a case ID shown to the citizen
- Office staff receives notification of new complaint

### US-05 – Acceptance Criteria

- Citizen can view case ID, submission date, status, and resolution notes
- Status updates in real-time as staff act on the complaint

### US-06 – Acceptance Criteria

- Form captures container serial number, type, and description of issue
- Submission creates a technician job ticket

- Citizen receives confirmation with expected visit window

**US-07 – Acceptance Criteria**

- Replacement request creates a technician delivery job
- System records $40 fee as collectible on delivery (cash or check)
- Technician can mark fee as collected upon delivery

**US-08 – Acceptance Criteria**

- Staff see a list of all open complaints with date, address, and status
- Staff can update status and add resolution notes
- Resolved complaints are archived but searchable

**US-09 – Acceptance Criteria**

- System supports 5 zones each assigned to a weekday
- Organics is scheduled weekly; garbage/recycling alternates every other week
- Schedule is visible to all relevant staff

**US-10 – Acceptance Criteria**

- Map displays all defined zones and routes
- Manager can draw/edit zone boundaries and route paths
- Changes are saved and reflected in scheduling immediately

**US-11 – Acceptance Criteria**

- Each route is assigned exactly one driver and one helper
- System prevents scheduling a crew member on two routes simultaneously
- Assignment history is retained

**US-12 – Acceptance Criteria**

- Manager can cancel or push a pickup by one day
- All registered citizens in affected zone receive a notification
- Schedule history is updated with reason for change

### US-13 – Acceptance Criteria

- Invoicing assistant can enter voucher number, date, vehicle, amount, and crew member name
- Duplicate voucher numbers are flagged
- Vouchers are searchable by truck or date range

### US-14 – Acceptance Criteria

- System compares gas station invoice line items to voucher records
- Unmatched items highlighted in red
- Matched total vs invoice total displayed for sign-off

### US-15 – Acceptance Criteria

- Dashboard shows total open, in-progress, and resolved complaints
- Drill-down to individual cases is available
- Data refreshes in real time

### US-16 – Acceptance Criteria

- Dashboard shows pending invoice total, matched vs unmatched vouchers
- Manager can mark invoice as approved for payment
- Approved invoices are locked from further editing

### US-17 – Acceptance Criteria

- Map shows each truck as a moving icon updated every 30 seconds
- Clicking a truck shows vehicle ID, driver, and current route
- Idle trucks (no movement for 15+ min) are highlighted

### US-18 – Acceptance Criteria

- Technician sees list of jobs for today sorted by address
- Each job shows type, citizen address, and container serial number
- Completed jobs show a green status indicator

### US-19 – Acceptance Criteria

- Technician can set status to: Assigned, In Progress, Completed, Unable to Complete
- Completion updates citizen portal in real time
- Unable to Complete requires a reason note

### US-20 – Acceptance Criteria

- Each container record includes serial number, type, address, issue date, and status
- Replacement creates a new record linked to the original
- Inventory count is visible to manager at a glance

# 10. Implementation Plan – Story to Task Mapping

Each user story from the product backlog is broken down into specific development and testing tasks below.

### US-01: Citizen Registration

| # | Task Description |
|---|---|
| 1 | Design citizen registration DB schema (name, address, email, phone) |
| 2 | Build registration form UI (web) |
| 3 | Implement server-side validation and duplicate email check |
| 4 | Generate inactive account record on submission |

### US-02: Email Verification

| # | Task Description |
|---|---|
| 1 | Generate unique token on registration |
| 2 | Send verification email with activation link |
| 3 | Implement link-click handler to activate account |
| 4 | Set token expiry (24 hours) and handle expired links |

### US-03: Pickup Notifications

| # | Task Description |
|---|---|
| 1 | Build notification preference settings (email/SMS/both) |
| 2 | Integrate email service (SMTP) and SMS gateway |
| 3 | Create scheduled job to send reminders 1 day before pickup |
| 4 | Trigger re-notification when holiday reschedule occurs |

### US-04: Missed Pickup Complaint

| # | Task Description |
|---|---|
| 1 | Design complaint form UI on citizen portal |

| | |
|---|---|
| 2 | Implement complaint submission API |
| 3 | Generate unique case ID and persist complaint record |
| 4 | Send notification to office staff on new complaint |

### US-05: Complaint Status Tracking

| # | Task Description |
|---|---|
| 1 | Build complaint detail view for citizen portal |
| 2 | Implement real-time status update query (polling or websocket) |
| 3 | Display case ID, date, status, and resolution notes |

### US-06: Container Maintenance Request

| # | Task Description |
|---|---|
| 1 | Design maintenance request form UI on citizen portal |
| 2 | Capture container serial number, type, and issue description |
| 3 | Create technician job ticket on submission |
| 4 | Send citizen confirmation with expected visit window |
| 5 | Link maintenance record to container serial number in inventory |

### US-07: Replacement Container Request

| # | Task Description |
|---|---|
| 1 | Add replacement option to container request form |
| 2 | Create technician delivery job ticket with $40 fee flag |
| 3 | Record fee as collectible on delivery (cash or check) |
| 4 | Build technician fee-collected confirmation UI |
| 5 | Update container inventory with new serial number on delivery |

### US-08: Complaint Management (Staff)

| # | Task Description |
|---|---|

| | |
|---|---|
| 1 | Build complaint list view with filters (open, in-progress, resolved) |
| 2 | Implement case detail view with status update and notes fields |
| 3 | Implement assignment of complaint to a staff member |
| 4 | Archive resolved complaints and keep them searchable |
| 5 | Send notification to citizen when complaint status changes |

### US-09: Pickup Scheduling

| # | Task Description |
|---|---|
| 1 | Design zone/route/schedule DB schema |
| 2 | Implement 5-zone, 5-weekday assignment logic |
| 3 | Build schedule management UI for office staff |
| 4 | Implement alternating garbage/recycling week logic |
| 5 | Implement organics as weekly pickup |

### US-10: Interactive Map (Zones & Routes)

| # | Task Description |
|---|---|
| 1 | Integrate map API (e.g., Google Maps/Leaflet) |
| 2 | Implement polygon drawing tool for zones |
| 3 | Implement polyline drawing tool for routes |
| 4 | Implement save/edit/delete for zones and routes |
| 5 | Link route to scheduled zone/day |

### US-11: Crew Assignment

| # | Task Description |
|---|---|
| 1 | Design crew assignment DB schema (driver, helper, route, date) |
| 2 | Build assignment UI for office staff |
| 3 | Implement conflict validation (no double-booking) |
| 4 | Implement assignment history logging |

### US-12: Holiday & Emergency Rescheduling

| # | Task Description |
|---|---|
| 1 | Build reschedule/cancel pickup UI for manager and scheduling assistant |
| 2 | Implement next-day auto-shift logic for holiday pickups |
| 3 | Allow manual cancellation or rescheduling of any zone/day |
| 4 | Trigger updated pickup notifications to all affected registered citizens |
| 5 | Log reason for change in schedule history |

### US-13 & 14: Fuel Voucher & Reconciliation

| # | Task Description |
|---|---|
| 1 | Design voucher DB schema (voucher#, date, truck, amount, crew) |
| 2 | Build voucher entry form for invoicing assistant |
| 3 | Implement duplicate voucher detection |
| 4 | Build invoice upload/entry for gas station bill |
| 5 | Implement line-by-line reconciliation engine |
| 6 | Build reconciliation dashboard with matched/unmatched highlighting |

### US-15: Complaints Dashboard (Manager)

| # | Task Description |
|---|---|
| 1 | Build complaints dashboard with open/in-progress/resolved counts |
| 2 | Implement drill-down to individual case detail from dashboard |
| 3 | Add average resolution time metric |
| 4 | Implement real-time data refresh (polling or websocket) |
| 5 | Filter dashboard by date range and complaint type |

### US-16: Financial Dashboard (Manager)

| # | Task Description |
|---|---|

| | |
|---|---|
| 1 | Build financial dashboard showing invoice period summary |
| 2 | Display matched vs. unmatched voucher totals |
| 3 | Implement invoice approval button (locks invoice from further editing) |
| 4 | Send pay order confirmation email to accounts manager on approval |
| 5 | Archive approved invoices with timestamp |

### US-17: GPS Fleet Tracking

| # | Task Description |
|---|---|
| 1 | Integrate GPS tracking API per truck |
| 2 | Build real-time fleet map view |
| 3 | Implement 30-second position refresh |
| 4 | Implement idle truck detection (15+ min threshold) and visual alert |

### US-18: Technician Job List

| # | Task Description |
|---|---|
| 1 | Build technician job list view sorted by address |
| 2 | Display job type, citizen address, container serial number, and issue description |
| 3 | Filter jobs by date (today's assignments) |
| 4 | Show completed jobs with green status indicator |
| 5 | Implement job detail view for full ticket information |

### US-19: Technician Job Status Update

| # | Task Description |
|---|---|
| 1 | Add status update control to job detail view (Assigned, In Progress, Completed, Unable to Complete) |
| 2 | Require reason note when marking Unable to Complete |
| 3 | Update citizen portal complaint/request status in real time on completion |
| 4 | Log timestamp and technician ID on each status change |

| | |
|---|---|
| 5 | Notify office staff when job is marked Unable to Complete |

### US-20: Container Inventory Management

| # | Task Description |
|---|---|
| 1 | Design container inventory DB schema (serial #, type, address, issue date, status) |
| 2 | Build inventory list view for manager with search and filter |
| 3 | Track replacement containers as new records linked to the original |
| 4 | Display total inventory count and breakdown by status on manager dashboard |
| 5 | Generate inventory report exportable by zone or date range |

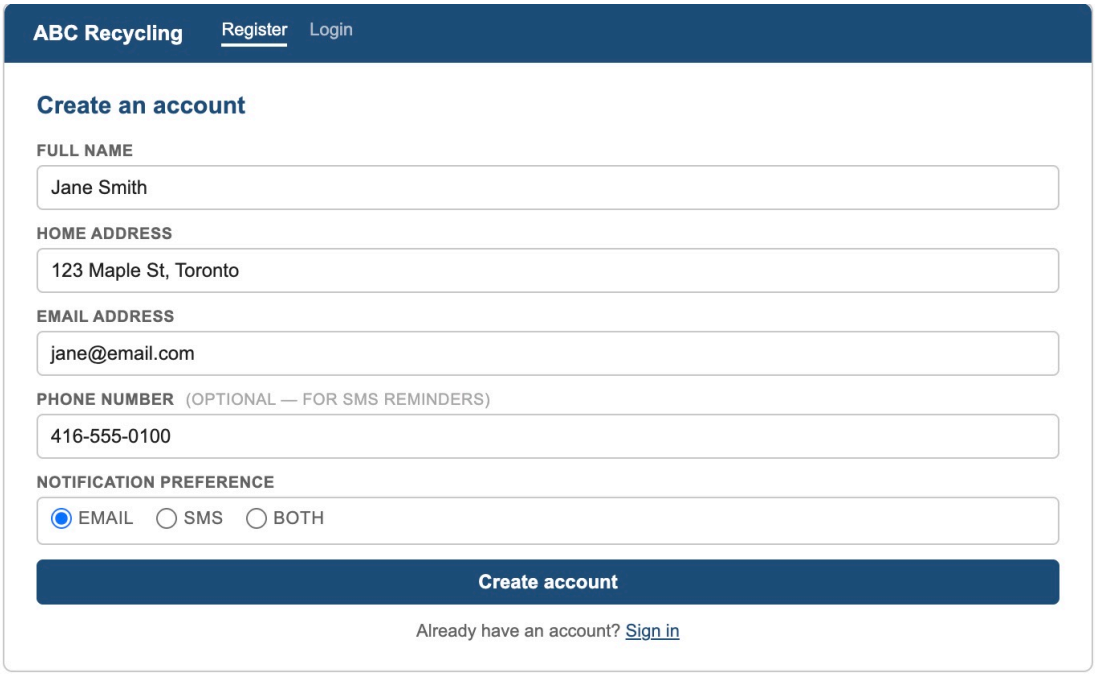
# 11. User Interface – Wireframes

The following section presents wireframes for all key screens of the ABC Recycling Management web application. The wireframes cover the complete system functionality and implement every user story across four portals: Citizen, Operations (Staff), Manager, and Technician.

## 11.1 Citizen Portal

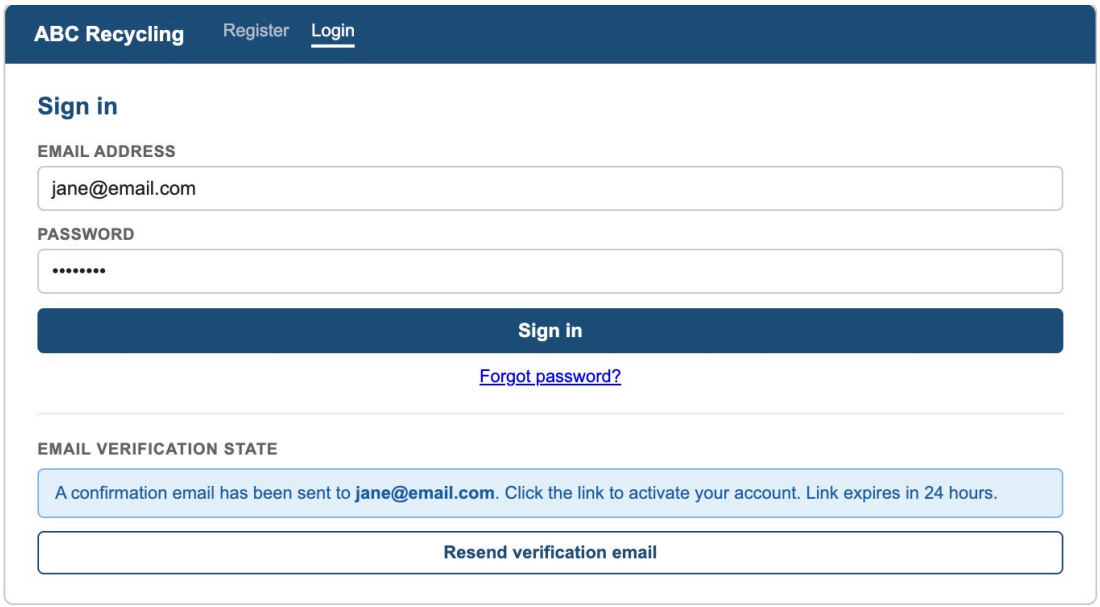
### Screen 1A: Citizen Registration

- Fields: Full Name, Address, Email, Phone (optional — for SMS reminders)
- Notification preference: Email, SMS, or Both
- Account remains inactive until email is verified



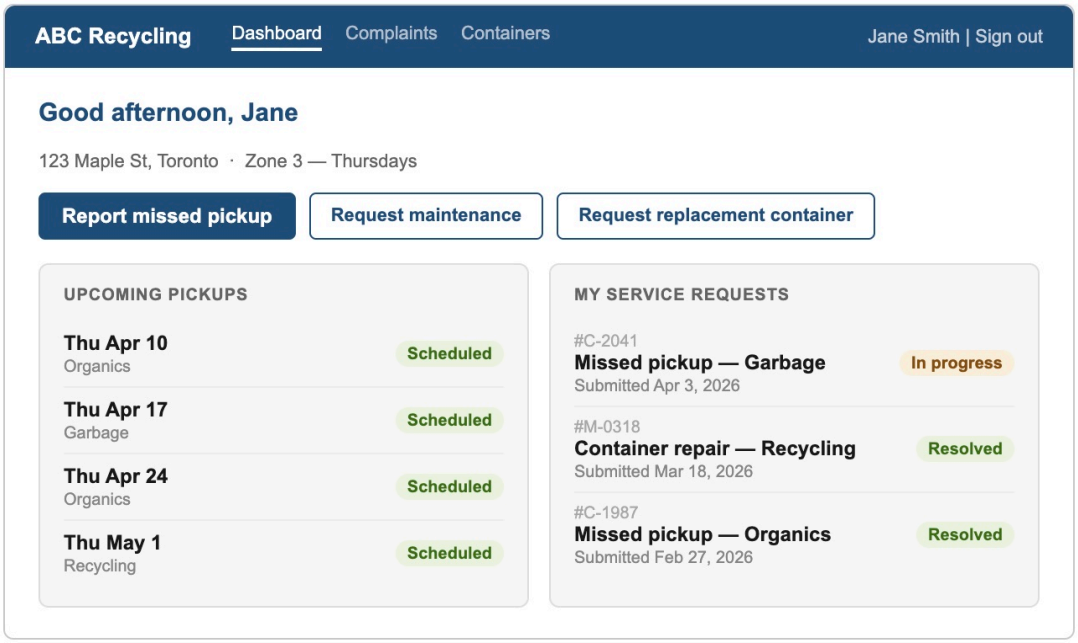

### Screen 1B: Login & Email Verification

- Login with email and password; Forgot Password link
- Email verification state shown after registration — link expires in 24 hours
- Resend verification email option available


ABC Recycling
Register
Login
Sign in
EMAIL ADDRESS
jane@email.com
PASSWORD
••••••••
Sign in
Forgot password?
EMAIL VERIFICATION STATE
A confirmation email has been sent to jane@email.com. Click the link to activate your account. Link expires in 24 hours.
Resend verification email

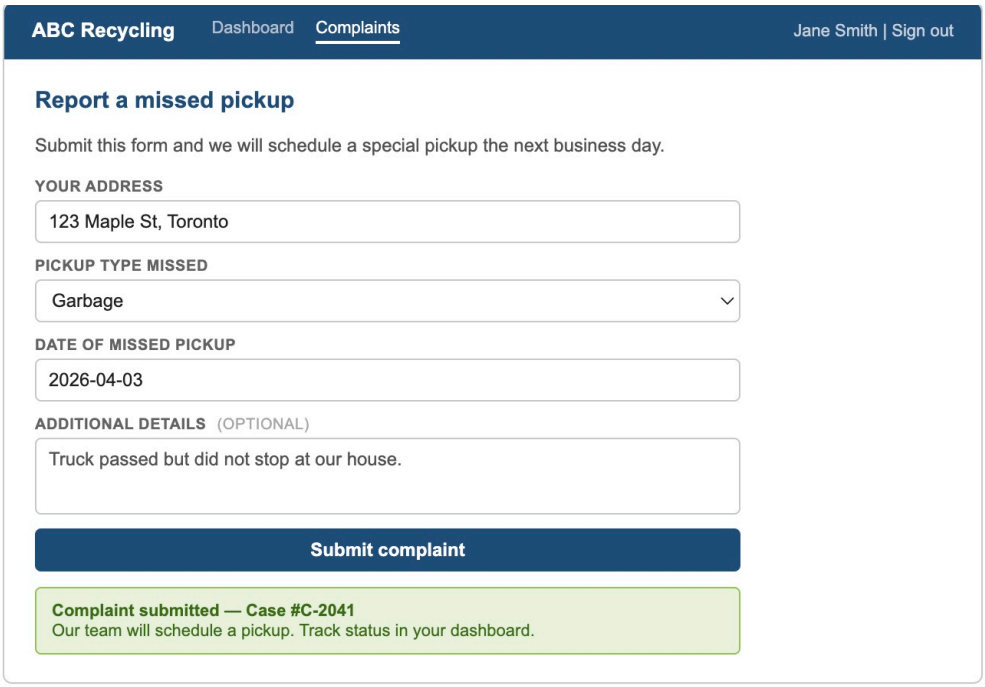
## Screen 2: Citizen Dashboard

- Upcoming pickup schedule for registered address (zone, day, waste type)
- Active service requests with status badges (Open, In Progress, Resolved)
- Quick action buttons: Report Missed Pickup | Request Maintenance | Request Replacement Container



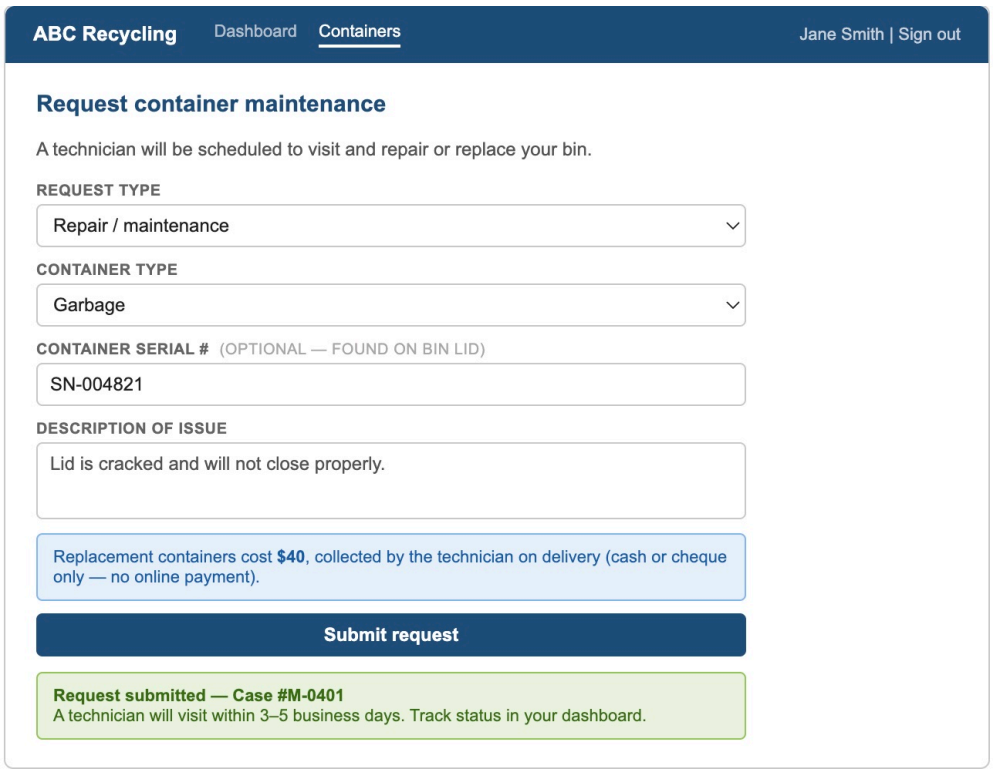
## Screen 3: File a Complaint / Missed Pickup

- Address pre-filled from citizen account
- Pickup type (Garbage/Recycling/Organics), date of missed pickup, and optional details
- Submission generates a unique Case ID shown to citizen; office staff notified automatically



## Screen 4: Container Maintenance Request

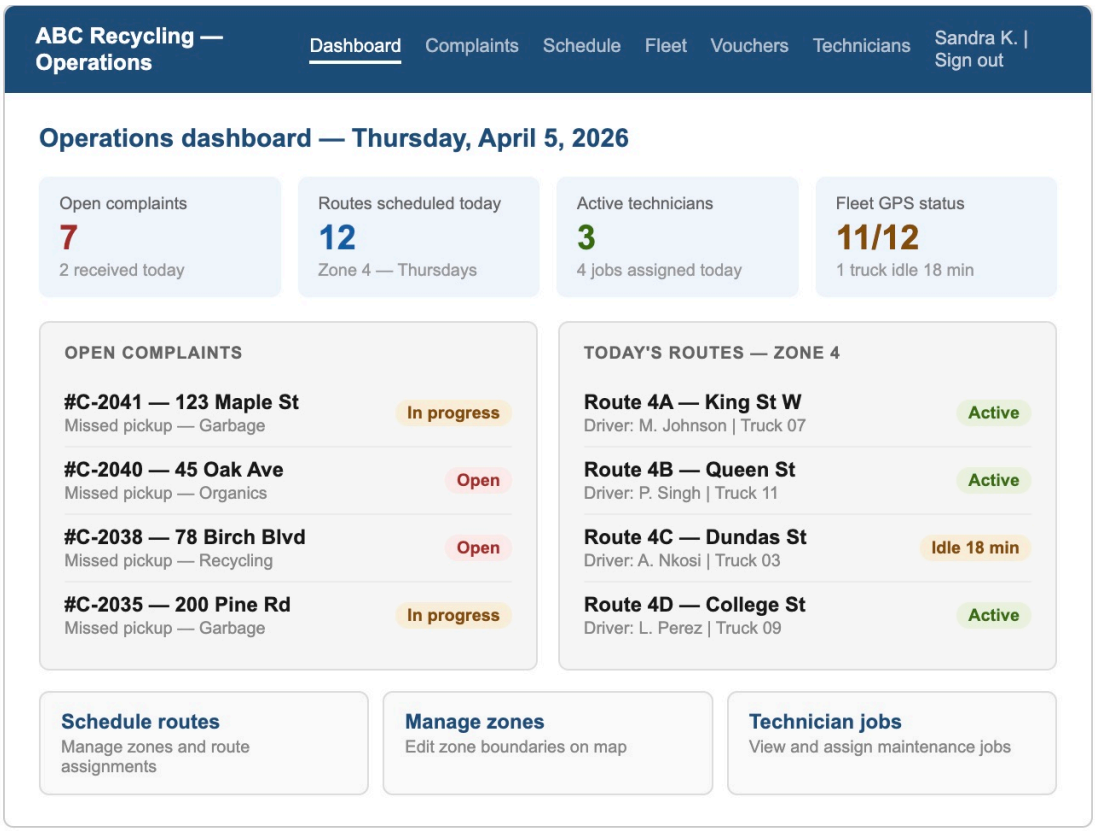
- Request type: Repair/Maintenance or Replacement Container ($40 fee)
- Container type, serial number (optional), and issue description
- Replacement containers cost $40, collected by technician on delivery — no online payment
- Submission creates a technician job ticket; citizen sees expected visit window



## 11.2 Office Staff / Operations Portal

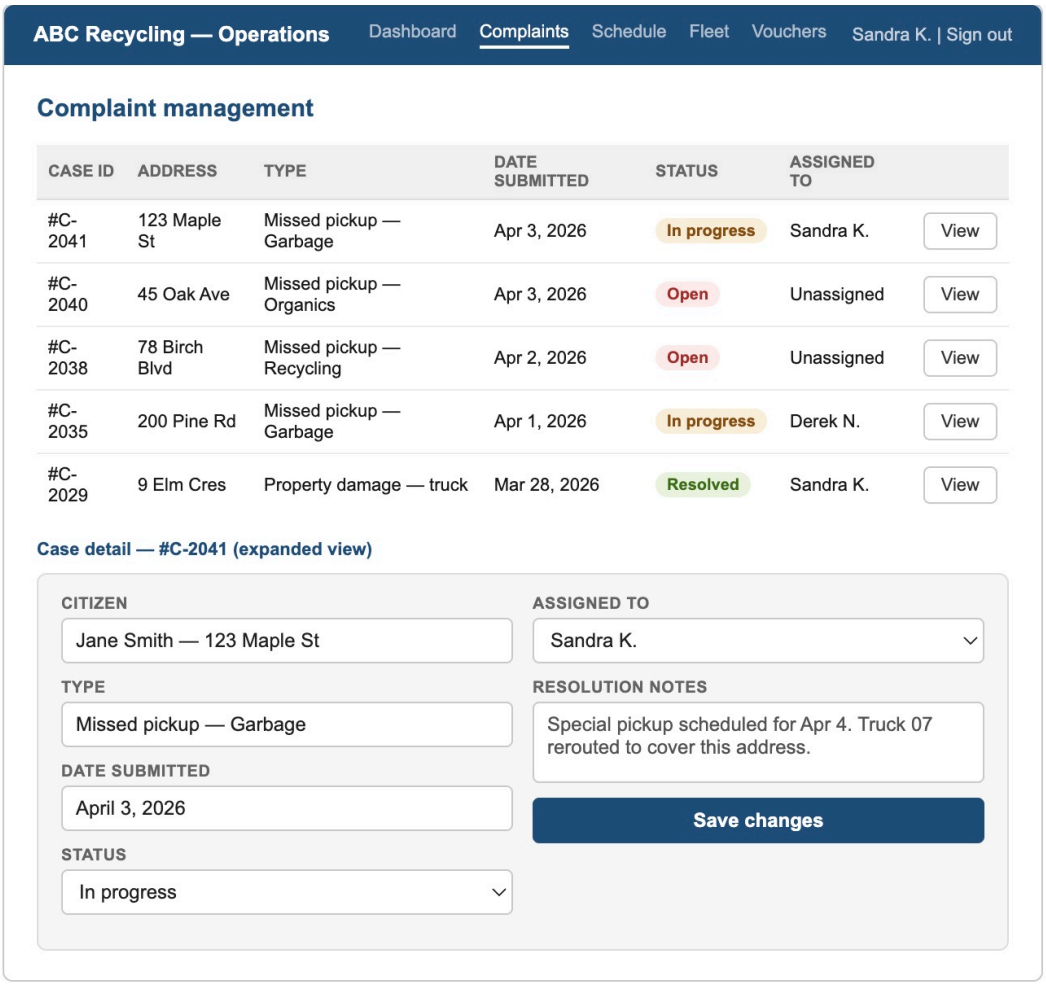
### Screen 5: Staff Dashboard (Home)

- Summary cards: Open Complaints, Routes Scheduled Today, Active Technicians, Fleet GPS Status
- Open complaints list and today's routes with real-time status indicators
- Quick access buttons: Schedule Routes, Manage Zones, Technician Jobs



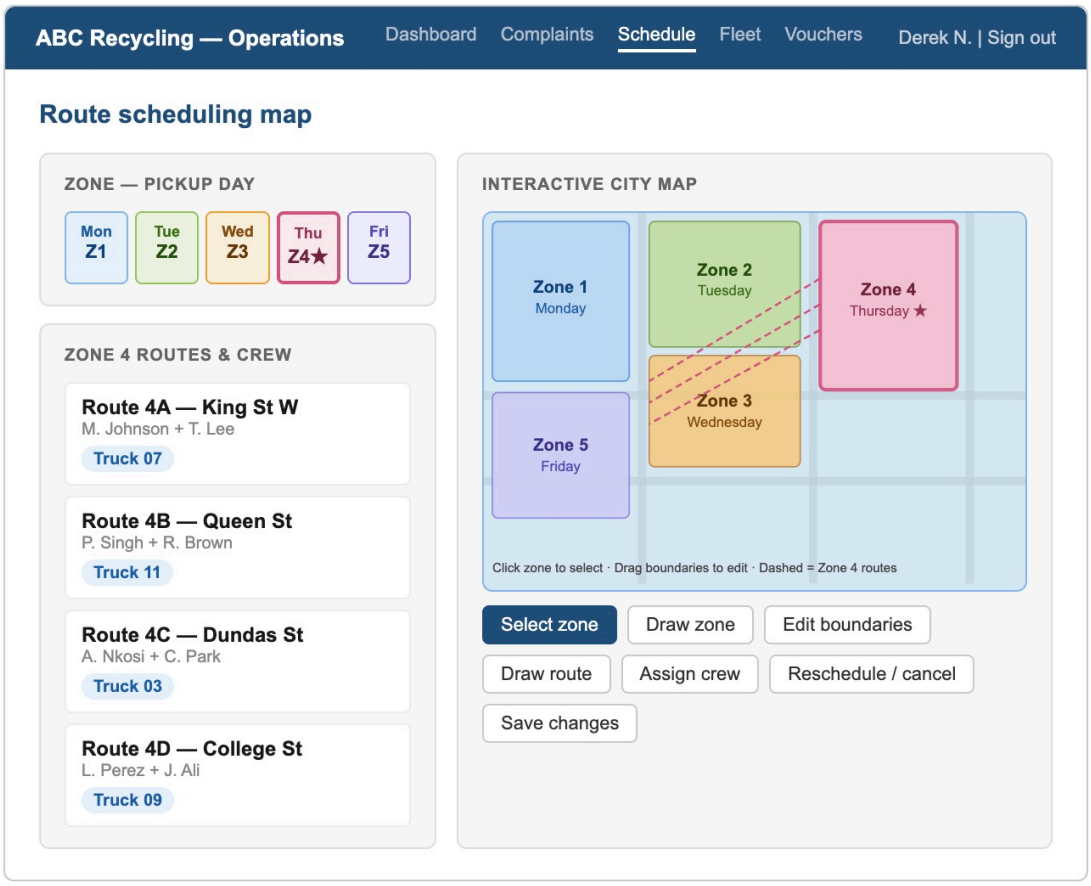
## Screen 6: Complaint Management

- Table: Case ID, Citizen Address, Type, Date Submitted, Status, Assigned Staff
- Click any row to open case detail view — update status, assign staff, add resolution notes
- Resolved complaints archived but remain searchable



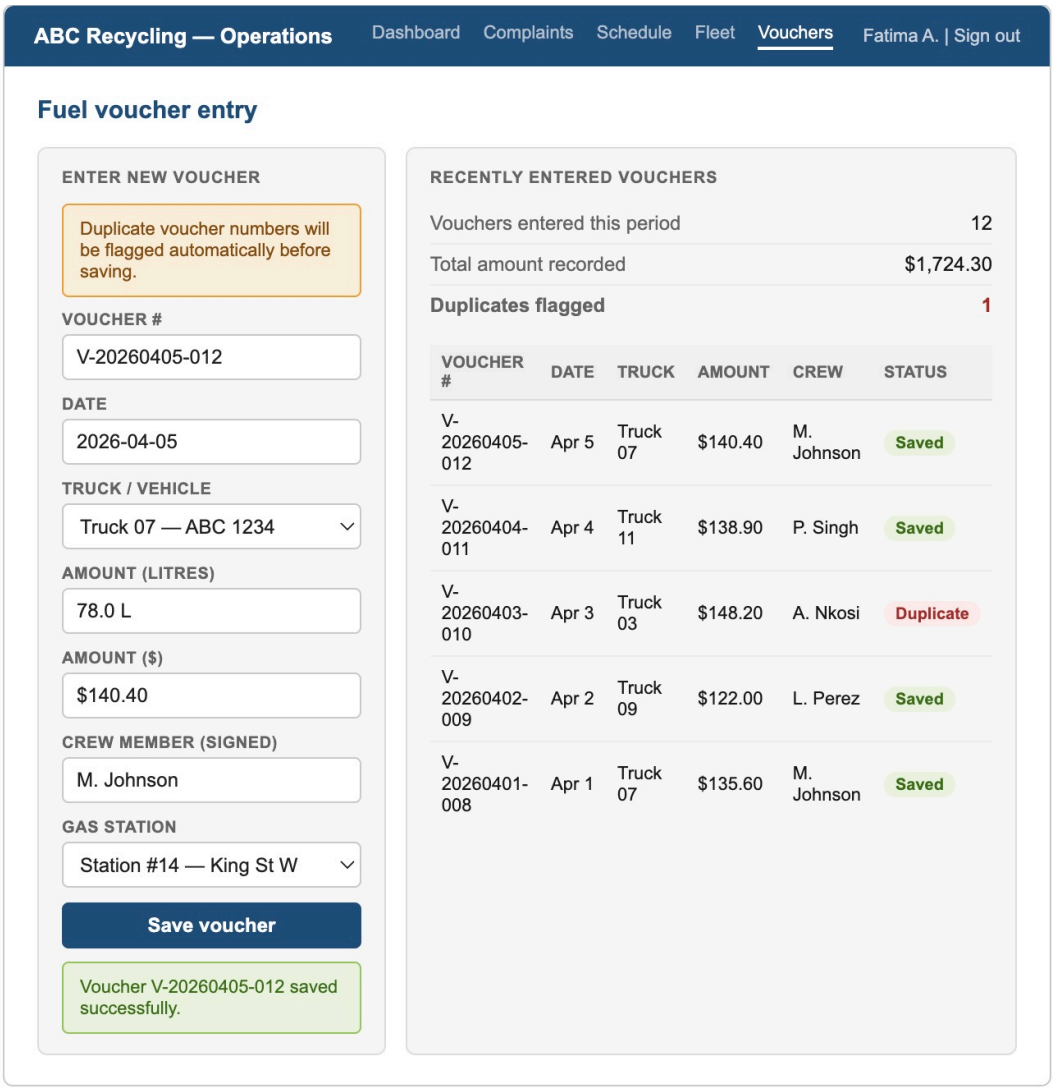
## Screen 7: Route Scheduling Map

- Zone-to-day assignment calendar (5 zones, Mon–Fri)
- Interactive city map: draw/edit zone boundaries, define routes per zone
- Zone 4 routes and crew assignments shown in side panel
- Tools: Select Zone, Draw Zone, Edit Boundaries, Draw Route, Assign Crew, Reschedule/Cancel, Save Changes



## Screen 8: Fuel Voucher Entry

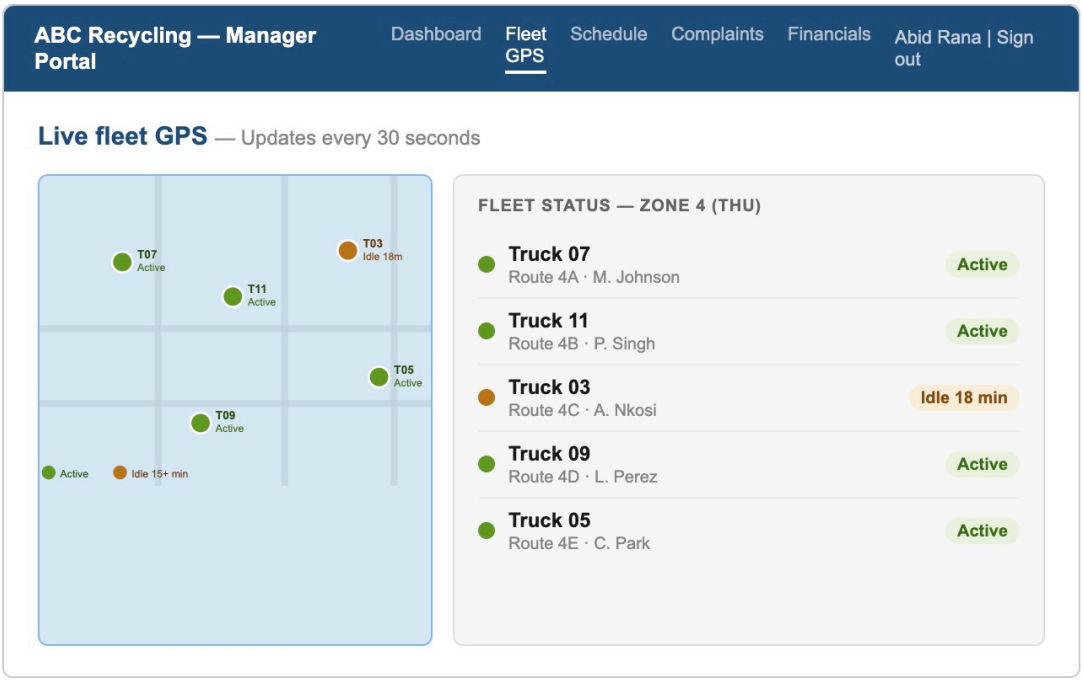
- Form: Voucher #, Date, Truck/Vehicle, Amount (litres and $), Crew Member, Gas Station
- Duplicate voucher numbers flagged automatically before saving
- List of recently entered vouchers with saved/duplicate status for verification against 15-day invoice



## 11.3 Manager Portal

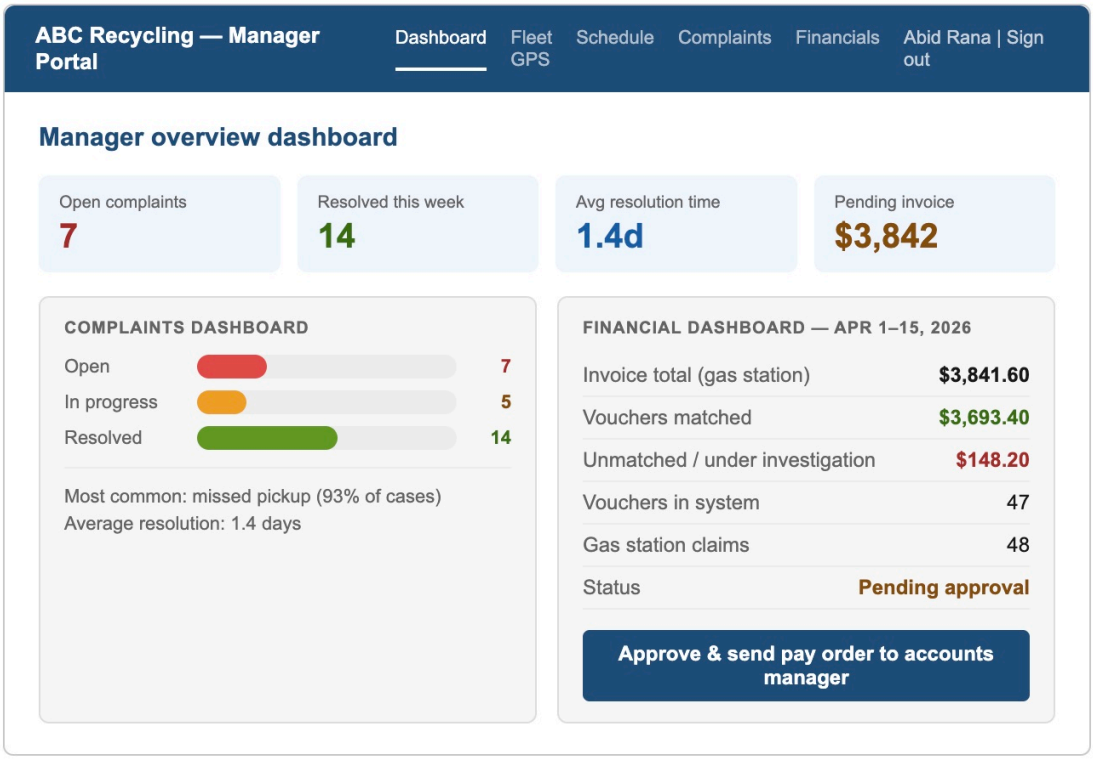
### Screen 9: GPS Fleet Map

- Live city map with truck positions updated every 30 seconds
- Green dot = active truck; Amber dot = idle 15+ minutes
- Fleet status panel shows each truck, route, and driver with status badge



## Screen 10: Manager Dashboards

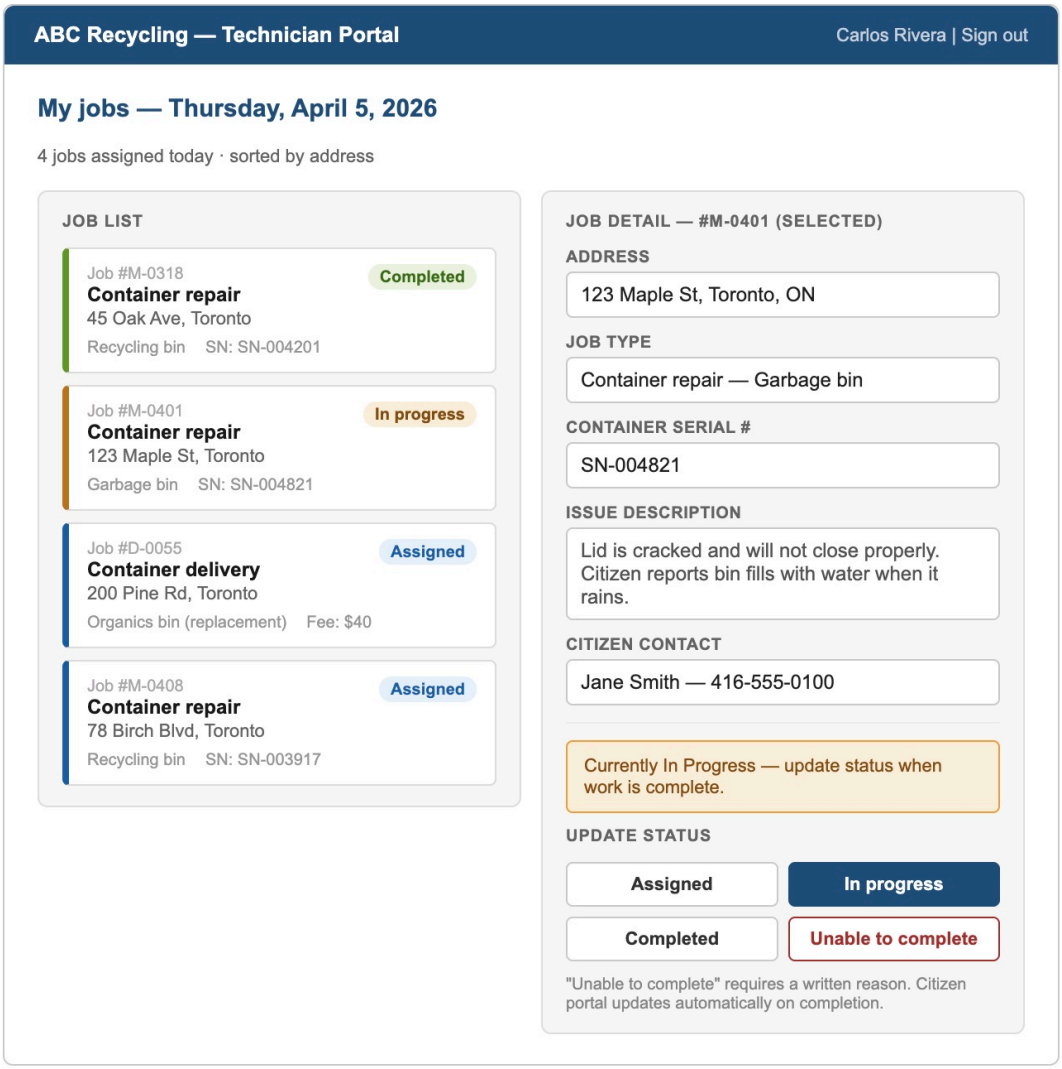
- Summary cards: Open Complaints, Resolved This Week, Avg Resolution Time, Pending Invoice
- Complaints Dashboard: bar chart of open/in-progress/resolved counts, most common type, avg resolution time
- Financial Dashboard: invoice total vs. matched vouchers, unmatched amount, approve button to send pay order to accounts manager



## 11.4 Technician Portal

### Screen 11: Technician Job List & Status Update

- Mobile-friendly list of all assigned jobs for the day, sorted by address
- Each job shows: type (maintenance/delivery), citizen address, container serial number, and issue description
- Job detail panel: address, job type, container serial #, issue description, citizen contact
- Status buttons: Assigned, In Progress, Completed, Unable to Complete
- Unable to Complete requires a written reason; citizen portal updates automatically on completion



# 12. Project Retrospective Report

## 12.1 Overview

This retrospective was conducted by the Group 49 SCRUM team at the conclusion of the ABC Recycling Management project. The purpose is to reflect on what went well, what could be improved, and what actionable improvements will be carried forward to future projects.

## 12.2 What Went Well

- The client interview (February 20, 2026) was highly productive. The team came prepared with specific, organized questions covering all functional areas, resulting in clear and detailed requirements.
- The use of the SCRUM framework kept the team organized and focused on deliverables with clear ownership.
- The SSR document provided a strong foundation for requirements, and the interview filled in critical gaps (staff roles, pickup frequency rules, payment handling, container serial number tracking).
- Team communication was consistent and respectful throughout the project.

## 12.3 Challenges Faced

- Coordinating schedules across team members for the client interview and internal meetings required flexibility.
- Some requirements were initially ambiguous (e.g., whether the accounts manager was a system user; whether a mobile app was needed). These were clarified in the interview.
- Translating broad SSR descriptions into precise, testable user stories required multiple rounds of refinement.

## 12.4 Lessons Learned

- Always prepare structured interview questions in advance — it significantly improves the quality and completeness of requirements gathered.
- Clarify role boundaries early (e.g., accounts manager is not a system actor) to prevent unnecessary scope.
- Document all client decisions (e.g., no mobile app, $40 cash/check for replacement, holiday = next-day pickup) immediately to avoid ambiguity later.
- Agile backlog prioritization is most effective when the team discusses business value collaboratively rather than estimating individually.

## 12.5 Agreed Actionable Improvements

- In future projects: send interview questions to the client 24 hours in advance so they can prepare more detailed answers.
- Use a shared real-time document (e.g., Google Docs) during interviews to record answers collaboratively instead of one person taking notes.
- Run a short backlog refinement session after every sprint to keep stories current with any evolving requirements.
- Create and maintain a glossary of domain terms (Zone, Route, Voucher, etc.) from day one to ensure consistent understanding across the team.

This retrospective formally concludes the Group 49 project for Comp2151 – Agile Software Development.

# 13. Conclusion

The ABC Recycling Management System is designed to improve operational efficiency, transparency, and citizen engagement for waste management services. By applying Agile SCRUM practices, the development team created a prioritized product backlog, clearly defined user stories, and measurable acceptance criteria. This structured approach ensures that the system focuses on the highest-value functionality first while remaining flexible to evolving requirements.

The proposed platform will allow ABC Recycling Management Inc. to manage scheduling, citizen service requests, container tracking, and fleet monitoring within a single integrated system. This will significantly improve operational coordination while providing citizens with a transparent and reliable method of interacting with their municipal waste management services.

Overall, the Agile approach used in this project supports ABC Recycling's ability to deliver high-quality service for its new municipal contract while positioning the company to expand operations to additional Canadian cities in the future.

***— End of Report —***